

Supplementary Information for MedEquiSeg: Shared Augmentation and Privileged-Prompt Reliability in Multimodal Medical Image Segmentation

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Canonical boundary and empty-mask metrics

All entries were recomputed after Protocol V3 predictions were restored to the native reference-mask grid with the same evaluator. A both-empty case receives perfect overlap and zero distance. A one-sided empty case receives zero overlap and infinite distance. For interpretability, the tables report finite-case HD95 and ASSD together

with the one-empty failure rate; finite distance values must not be interpreted without the corresponding failure rate. NSD uses a two-pixel tolerance.

Table S1 Canonical Protocol V3 boundary and empty-mask metrics for BUSI. Values are mean $\pm$ sample standard deviation over seeds 123, 456, and 789. NSD and failure rate are percentages. HD95 and ASSD are finite-case means in native-resolution pixels; they must be interpreted jointly with the one-empty failure rate. A dagger marks image-only methods.

Method	NSD (%)	finite HD95 (px)	finite ASSD (px)	one-empty (%)
MedCLIPSeg	41.11 $\pm$ 0.66	32.64 $\pm$ 6.09	10.92 $\pm$ 1.67	0.00 $\pm$ 0.00
CausalCLIPSeg	43.58 $\pm$ 0.50	40.61 $\pm$ 3.01	12.55 $\pm$ 0.97	0.00 $\pm$ 0.00
LViT	41.26 $\pm$ 1.45	48.09 $\pm$ 2.87	14.39 $\pm$ 0.67	1.28 $\pm$ 1.28
MEDEquISEG (Ours)	43.96 $\pm$ 1.30	35.86 $\pm$ 3.44	11.42 $\pm$ 0.92	0.00 $\pm$ 0.00
RollingUNet [†]	29.81 $\pm$ 2.29	65.72 $\pm$ 2.14	20.51 $\pm$ 1.38	7.26 $\pm$ 1.96
U-Net [†]	12.27 $\pm$ 1.86	120.79 $\pm$ 31.04	30.80 $\pm$ 6.69	16.67 $\pm$ 0.00
U-Net++ [†]	34.69 $\pm$ 1.46	48.76 $\pm$ 3.25	15.78 $\pm$ 0.46	6.41 $\pm$ 1.28
nnU-Net [†]	33.76 $\pm$ 0.52	44.65 $\pm$ 2.83	15.43 $\pm$ 2.08	8.55 $\pm$ 1.96
U-Mamba [†]	35.48 $\pm$ 1.47	49.16 $\pm$ 12.17	16.64 $\pm$ 2.79	6.84 $\pm$ 0.74
U-KAN [†]	37.10 $\pm$ 2.93	48.21 $\pm$ 2.73	16.79 $\pm$ 1.73	5.98 $\pm$ 2.96

Table S2 Canonical Protocol V3 boundary and empty-mask metrics for ClinicDB. Values are mean $\pm$ sample standard deviation over seeds 123, 456, and 789. NSD and failure rate are percentages. HD95 and ASSD are finite-case means in native-resolution pixels; they must be interpreted jointly with the one-empty failure rate. A dagger marks image-only methods.

Method	NSD (%)	finite HD95 (px)	finite ASSD (px)	one-empty (%)
MedCLIPSeg	58.50 $\pm$ 1.19	12.25 $\pm$ 1.07	3.40 $\pm$ 0.15	0.00 $\pm$ 0.00
CausalCLIPSeg	64.26 $\pm$ 2.37	15.78 $\pm$ 1.04	3.84 $\pm$ 0.28	0.00 $\pm$ 0.00
LViT	65.26 $\pm$ 1.48	16.28 $\pm$ 3.10	3.96 $\pm$ 0.71	0.00 $\pm$ 0.00
MEDEquISEG (Ours)	67.00 $\pm$ 0.24	14.99 $\pm$ 1.08	3.61 $\pm$ 0.34	0.00 $\pm$ 0.00
RollingUNet [†]	49.88 $\pm$ 2.32	21.46 $\pm$ 1.76	5.19 $\pm$ 0.18	1.09 $\pm$ 0.95
U-Net [†]	38.83 $\pm$ 3.45	36.13 $\pm$ 5.40	8.85 $\pm$ 1.41	0.00 $\pm$ 0.00
U-Net++ [†]	60.07 $\pm$ 1.13	18.47 $\pm$ 1.58	4.75 $\pm$ 0.20	0.00 $\pm$ 0.00
nnU-Net [†]	70.58 $\pm$ 0.48	9.82 $\pm$ 1.23	2.71 $\pm$ 0.25	0.00 $\pm$ 0.00
U-Mamba [†]	69.79 $\pm$ 1.25	13.23 $\pm$ 0.52	3.60 $\pm$ 0.22	0.00 $\pm$ 0.00
U-KAN [†]	62.93 $\pm$ 0.33	15.17 $\pm$ 0.94	3.89 $\pm$ 0.18	0.00 $\pm$ 0.00

Table S3 Canonical Protocol V3 boundary and empty-mask metrics for BUS-BRA. Values are mean $\pm$ sample standard deviation over seeds 123, 456, and 789. NSD and failure rate are percentages. HD95 and ASSD are finite-case means in native-resolution pixels; they must be interpreted jointly with the one-empty failure rate. A dagger marks image-only methods.

Method	NSD (%)	finite HD95 (px)	finite ASSD (px)	one-empty (%)
MedCLIPSeg	45.54 $\pm$ 0.76	16.13 $\pm$ 1.54	5.09 $\pm$ 0.66	0.00 $\pm$ 0.00
CausalCLIPSeg	49.57 $\pm$ 1.27	15.53 $\pm$ 0.67	4.99 $\pm$ 0.42	0.00 $\pm$ 0.00
LViT	50.40 $\pm$ 0.61	20.00 $\pm$ 0.82	5.84 $\pm$ 0.30	0.00 $\pm$ 0.00
MEDEquISEG (Ours)	52.53 $\pm$ 0.29	13.81 $\pm$ 0.59	4.31 $\pm$ 0.15	0.00 $\pm$ 0.00
RollingUNet [†]	50.67 $\pm$ 1.21	18.71 $\pm$ 1.34	5.51 $\pm$ 0.35	0.12 $\pm$ 0.20
U-Net [†]	44.75 $\pm$ 0.66	22.17 $\pm$ 0.69	6.05 $\pm$ 0.11	0.00 $\pm$ 0.00
U-Net++ [†]	53.04 $\pm$ 0.20	18.50 $\pm$ 0.52	5.22 $\pm$ 0.05	0.00 $\pm$ 0.00
nnU-Net [†]	53.89 $\pm$ 0.41	14.83 $\pm$ 0.61	4.52 $\pm$ 0.19	0.00 $\pm$ 0.00
U-Mamba [†]	55.25 $\pm$ 0.21	13.72 $\pm$ 0.74	4.32 $\pm$ 0.26	0.00 $\pm$ 0.00
U-KAN [†]	52.63 $\pm$ 0.68	16.98 $\pm$ 0.32	5.19 $\pm$ 0.29	0.00 $\pm$ 0.00

Table S4 Canonical Protocol V3 boundary and empty-mask metrics for BRISC. Values are mean $\pm$ sample standard deviation over seeds 123, 456, and 789. NSD and failure rate are percentages. HD95 and ASSD are finite-case means in native-resolution pixels; they must be interpreted jointly with the one-empty failure rate. A dagger marks image-only methods.

Method	NSD (%)	finite HD95 (px)	finite ASSD (px)	one-empty (%)
MedCLIPSeg	48.90 $\pm$ 4.84	15.25 $\pm$ 0.91	5.82 $\pm$ 0.70	0.90 $\pm$ 0.40
CausalCLIPSeg	65.90 $\pm$ 1.43	14.69 $\pm$ 1.46	4.82 $\pm$ 0.32	0.49 $\pm$ 0.75
LViT	70.78 $\pm$ 0.45	14.82 $\pm$ 0.50	4.37 $\pm$ 0.06	1.64 $\pm$ 0.34
MEDEquISEG (Ours)	67.12 $\pm$ 0.16	12.63 $\pm$ 0.20	4.23 $\pm$ 0.30	0.37 $\pm$ 0.17
RollingUNet [†]	64.64 $\pm$ 0.72	13.60 $\pm$ 0.24	4.67 $\pm$ 0.25	1.64 $\pm$ 0.36
U-Net [†]	61.68 $\pm$ 0.98	13.82 $\pm$ 0.31	4.87 $\pm$ 0.11	0.64 $\pm$ 0.23
U-Net++ [†]	67.66 $\pm$ 0.47	13.55 $\pm$ 0.16	4.75 $\pm$ 0.17	0.75 $\pm$ 0.32
nnU-Net [†]	73.33 $\pm$ 0.51	12.58 $\pm$ 0.54	3.95 $\pm$ 0.26	1.49 $\pm$ 0.39
U-Mamba [†]	70.19 $\pm$ 1.10	13.23 $\pm$ 0.30	4.62 $\pm$ 0.18	2.02 $\pm$ 0.30
U-KAN [†]	64.89 $\pm$ 0.53	13.02 $\pm$ 0.69	4.28 $\pm$ 0.18	1.57 $\pm$ 0.40

Table S5 Canonical Protocol V3 boundary and empty-mask metrics for COVID-19. Values are mean $\pm$ sample standard deviation over seeds 123, 456, and 789. NSD and failure rate are percentages. HD95 and ASSD are finite-case means in native-resolution pixels; they must be interpreted jointly with the one-empty failure rate. A dagger marks image-only methods.

Method	NSD (%)	finite HD95 (px)	finite ASSD (px)	one-empty (%)
MedCLIPSeg	67.37 $\pm$ 2.09	18.10 $\pm$ 3.18	4.45 $\pm$ 0.65	0.00 $\pm$ 0.00
CausalCLIPSeg	68.24 $\pm$ 0.19	20.02 $\pm$ 0.13	5.50 $\pm$ 0.09	0.08 $\pm$ 0.07
LViT	65.75 $\pm$ 0.54	27.13 $\pm$ 0.55	6.96 $\pm$ 0.15	0.65 $\pm$ 0.36
MEDEquISEG (Ours)	71.08 $\pm$ 0.26	15.03 $\pm$ 0.71	3.72 $\pm$ 0.18	0.06 $\pm$ 0.03
RollingUNet [†]	66.47 $\pm$ 0.36	29.17 $\pm$ 0.76	6.86 $\pm$ 0.09	0.19 $\pm$ 0.09
U-Net [†]	64.08 $\pm$ 0.45	30.47 $\pm$ 0.20	7.13 $\pm$ 0.08	0.00 $\pm$ 0.00
U-Net++ [†]	66.57 $\pm$ 0.15	28.02 $\pm$ 0.24	6.53 $\pm$ 0.05	0.00 $\pm$ 0.00
nnU-Net [†]	66.09 $\pm$ 0.31	28.01 $\pm$ 0.65	6.55 $\pm$ 0.16	0.71 $\pm$ 0.14
U-Mamba [†]	66.35 $\pm$ 0.43	27.78 $\pm$ 0.73	6.70 $\pm$ 0.18	0.62 $\pm$ 0.16
U-KAN [†]	67.23 $\pm$ 0.31	29.32 $\pm$ 1.00	6.88 $\pm$ 0.23	0.38 $\pm$ 0.17

COVID-19 subject-cluster sensitivity

This post hoc sensitivity analysis uses the reported frozen checkpoints. It accounts for repeated test images from 474 recoverable subjects, but it does not repair the known overlap of recoverable subject IDs between the fixed training and validation folders.

Table S6 COVID-19 subject-cluster sensitivity under the original Protocol V3 test set. The image-weighted delta is the main evaluator point estimate, and its 95% interval resamples recoverable subjects and training seeds with 10,000 replicates. Subject-weighted deltas first average the three-seed difference within each of 474 subjects and then weight subjects equally. p_{subj} is the two-sided Wilcoxon test on those equal-subject mean differences. Thus, the confidence interval and p -value describe explicitly different estimands. All effects are percentage points.

Comparison	Metric	Image Δ	Cluster 95% CI	Subject Δ	p_{subj}
MEDEQUISEG – w/o Rewrite	Dice	−0.16	[−0.48, 0.14]	−0.21	.006
	IoU	−0.18	[−0.52, 0.15]	−0.28	.005
MEDEQUISEG – U-Net++	Dice	+4.08	[3.48, 4.72]	+5.30	< .001
	IoU	+5.00	[4.30, 5.68]	+5.95	< .001
True – shuffled	Dice	+15.29	[13.99, 16.66]	+17.68	< .001
	IoU	+18.04	[16.61, 19.51]	+19.39	< .001

Realized rewrite-trigger audit

The following table replays the exact deterministic MEDEQUISEG augmentation sampler without loading images. It reports realized rather than analytic trigger rates.

Table S7 Realized MEDEQUISEG rewrite-trigger audit over all training cases, 100 epochs, and seeds 123, 456, and 789. Spatial coverage is the percentage of training prompts containing a recognized direction phrase. Discrete-plan rate counts samples receiving at least one flip or right-angle rotation; string-change rate counts samples whose final prompt differs from its input after the complete composed rewrite. Values for event rates are mean±sample standard deviation across seeds.

Dataset	Train	Events/seed	Spatial (%)	Discrete plan (%)	String change (%)
BUSI	624	62400	82.85	48.42 ± 0.03	20.27 ± 0.10
ClinicDB	490	49000	100.00	48.37 ± 0.24	28.48 ± 0.21
BUS-BRA	1311	131100	100.00	48.42 ± 0.20	12.44 ± 0.12
BRISC	3881	388100	78.23	48.49 ± 0.12	5.75 ± 0.04
COVID-19	5716	571600	100.00	48.54 ± 0.04	45.75 ± 0.06

Mask-presence sensitivity

This analysis separates lesion-containing cases from empty-ground-truth cases so that correct empty predictions do not implicitly stand in for lesion-boundary quality.

Table S8 Mask-presence sensitivity analysis under canonical Protocol V3. Non-empty Dice and IoU are computed only on test images containing foreground. Empty specificity is the percentage of empty-ground-truth images predicted as empty. Values are mean $\pm$ sample standard deviation over seeds 123, 456, and 789. A dash indicates that the test split contains no empty masks. The image-only reference is selected by the highest overall three-seed Dice within each dataset before this subgroup analysis.

Dataset	Method	Dice	IoU	Empty specificity
BUSI	MEDEQUISEG (Ours)	82.94 $\pm$ 0.55	75.39 $\pm$ 0.28	100.00 $\pm$ 0.00
	MedCLIPSeg	83.36 $\pm$ 1.30	75.22 $\pm$ 1.34	100.00 $\pm$ 0.00
	U-KAN [†]	79.70 $\pm$ 0.48	71.63 $\pm$ 0.57	64.10 $\pm$ 17.76
ClinicDB	MEDEQUISEG (Ours)	92.07 $\pm$ 0.71	86.36 $\pm$ 0.61	–
	MedCLIPSeg	91.96 $\pm$ 0.07	85.64 $\pm$ 0.10	–
	nnU-Net [†]	93.62 $\pm$ 0.31	88.63 $\pm$ 0.29	–
BUS-BRA	MEDEQUISEG (Ours)	90.24 $\pm$ 0.14	83.15 $\pm$ 0.10	–
	MedCLIPSeg	88.47 $\pm$ 0.66	80.46 $\pm$ 0.76	–
	U-Mamba [†]	90.34 $\pm$ 0.46	83.58 $\pm$ 0.47	–
BRISC	MEDEQUISEG (Ours)	84.82 $\pm$ 0.17	76.68 $\pm$ 0.16	100.00 $\pm$ 0.00
	MedCLIPSeg	77.92 $\pm$ 2.12	67.36 $\pm$ 2.67	100.00 $\pm$ 0.00
	nnU-Net [†]	85.08 $\pm$ 0.34	77.99 $\pm$ 0.43	99.28 $\pm$ 0.72
COVID-19	MEDEQUISEG (Ours)	84.08 $\pm$ 0.13	75.51 $\pm$ 0.14	–
	MedCLIPSeg	82.62 $\pm$ 1.19	73.47 $\pm$ 1.48	–
	U-Net++ [†]	80.00 $\pm$ 0.06	70.51 $\pm$ 0.08	–

[†]Image-only contextual reference; cross-modality differences are descriptive because public benchmark prompts contain target-derived attributes.

Complete component ablation

The main manuscript reports the compact Public-5 component summary. The full dataset-level matrix is retained here to show that the macro improvement is not uniform across datasets and to preserve all 90 ordered-ablation cells.

Table S9 Complete frozen follow-up three-seed ordered component ablation on the fixed V3 data and evaluation protocol. ATConv denotes the two forward-active projector replacements; the shared-plan rows use the separately frozen R11 training-policy amendment. Values are Dice and IoU percentages (mean $\pm$ sample standard deviation). Bold and underline identify the best and second-best configuration within each dataset and metric.

Configuration	Metric	BUSI	ClinicDB	BUS-BRA
Base	Dice	84.50 $\pm$ 0.82	90.15 $\pm$ 0.97	89.26 $\pm$ 0.14
	IoU	77.86 $\pm$ 0.80	84.19 $\pm$ 0.91	81.83 $\pm$ 0.22
+ BioMedCLIP	Dice	83.76 $\pm$ 0.47	90.47 $\pm$ 1.03	89.17 $\pm$ 0.17
	IoU	76.96 $\pm$ 0.59	84.88 $\pm$ 0.96	81.67 $\pm$ 0.22
+ ATConv	Dice	84.76 $\pm$ 0.48	90.87 $\pm$ 0.99	89.19 $\pm$ 0.42
	IoU	78.08 $\pm$ 0.26	84.88 $\pm$ 1.17	81.78 $\pm$ 0.53
+ BioMedCLIP + ATConv	Dice	84.87 $\pm$ 0.46	91.49 $\pm$ 0.47	89.08 $\pm$ 0.22
	IoU	78.08 $\pm$ 0.51	85.57 $\pm$ 0.56	81.76 $\pm$ 0.23
+ Shared plan (no rewrite)	Dice	85.60 $\pm$ 0.58	92.37 $\pm$ 0.93	90.14 $\pm$ 0.16
	IoU	<u>79.36 $\pm$ 0.63</u>	86.82 $\pm$ 1.11	<u>83.03 $\pm$ 0.18</u>
MedEquiSeg (Ours)	Dice	85.78 $\pm$ 0.45	92.07 $\pm$ 0.71	90.24 $\pm$ 0.14
	IoU	79.49 $\pm$ 0.23	<u>86.36 $\pm$ 0.61</u>	83.15 $\pm$ 0.10
Configuration	Metric	BRISC	COVID-19	Public-5
Base	Dice	86.52 $\pm$ 0.17	81.65 $\pm$ 0.08	86.41 $\pm$ 0.35
	IoU	79.52 $\pm$ 0.14	72.71 $\pm$ 0.14	79.22 $\pm$ 0.34
+ BioMedCLIP	Dice	86.58 $\pm$ 0.06	83.35 $\pm$ 0.07	86.67 $\pm$ 0.08
	IoU	79.60 $\pm$ 0.04	74.49 $\pm$ 0.09	79.52 $\pm$ 0.06
+ ATConv	Dice	86.34 $\pm$ 0.11	82.02 $\pm$ 0.16	86.63 $\pm$ 0.32
	IoU	79.31 $\pm$ 0.14	73.06 $\pm$ 0.11	79.42 $\pm$ 0.31
+ BioMedCLIP + ATConv	Dice	86.39 $\pm$ 0.25	83.32 $\pm$ 0.10	87.03 $\pm$ 0.15
	IoU	79.41 $\pm$ 0.18	74.42 $\pm$ 0.19	79.85 $\pm$ 0.23
+ Shared plan (no rewrite)	Dice	87.46 $\pm$ 0.20	84.24 $\pm$ 0.13	87.96 $\pm$ 0.05
	IoU	80.59 $\pm$ 0.20	75.69 $\pm$ 0.13	81.10 $\pm$ 0.08
MedEquiSeg (Ours)	Dice	87.18 $\pm$ 0.14	84.08 $\pm$ 0.13	87.87 $\pm$ 0.17
	IoU	<u>80.32 $\pm$ 0.13</u>	<u>75.51 $\pm$ 0.14</u>	<u>80.97 $\pm$ 0.13</u>

COVID-19 grouped train-validation sensitivity

The fixed public test set is unchanged in this analysis. Recoverable subject tokens are assigned to only one training or validation split; rows without a recoverable token remain case-level groups. This is a split-sensitivity analysis, not evidence that every image in the source release has a known patient identity.

Table S10 COVID-19 grouped train-validation sensitivity. The fixed test set is unchanged. The sensitivity manifest contains 5,706 training, 1,439 validation, and 2,113 test images; overlap among 2,130 recoverable subject tokens is zero across all splits. Values are percentages (mean $\pm$ sample standard deviation over seeds 123, 456, and 789). Δ is the grouped-validation result minus a separately trained, configuration-matched original-split sensitivity result in percentage points; these sensitivity checkpoints are distinct from the frozen ordered-ablation checkpoint set in the main table.

Method	Dice	Original	Δ	IoU	Original	Δ
MEDEQUISEG	83.99 $\pm$ 0.07	84.18 $\pm$ 0.13	-0.19	75.35 $\pm$ 0.09	75.64 $\pm$ 0.14	-0.28
No rewrite	84.28 $\pm$ 0.06	84.28 $\pm$ 0.10	+0.00	75.71 $\pm$ 0.10	75.77 $\pm$ 0.15	-0.06
U-Net++	80.14 $\pm$ 0.09	80.00 $\pm$ 0.05	+0.14	70.68 $\pm$ 0.18	70.51 $\pm$ 0.10	+0.17

Within the grouped-validation checkpoint set, the direct MEDEQUISEG minus MEDEQUISEG w/o Prompt Rewrite Dice contrast is -0.29 percentage points (paired-case 95% confidence interval -0.55 to -0.03). After equal-subject averaging over 474 recoverable subjects, the effect is -0.40 points; the subject-cluster interval is -0.58 to -0.02 and the two-sided Wilcoxon p -value is $< .001$. The grouped sensitivity therefore does not rescue an accuracy-benefit claim for rewriting.

Protocol and implementation verification

This audit distinguishes the original V3 data/evaluation lock from the later shared-plan training amendment and verifies the effective ATConv execution graph. It is descriptive of the runs already reported and does not alter checkpoints or predictions.

Table S11 Protocol and implementation verification performed before manuscript revision. Hashes are abbreviated for display; complete values and machine-readable audit outputs are included in the results archive.

Audit item	Executed evidence	Manuscript treatment
V3 lock scope	Protocol hash abb8ccb2 ... fixes manifests, split roles, mask rules, seeds, threshold, and the evaluation specification; evaluator source identity is carried separately by the code digest. It does not contain the full model or augmentation policy.	Used as a data/evaluation identifier, not a complete training-policy lock.
Shared-plan routing	Original V3 routing is horizontal-flip-only for ClinicDB and COVID-19 and excludes continuous geometry. The reported R11 sampler applies multi-geometry plus mild affine, translation, and zoom to every dataset.	Reported as a separately frozen follow-up amendment.
Rewrite pairing	The rewrite and no-rewrite run names are normalized to the same keyed sampler seed, using training seed, epoch, and case identifier.	Accepted as an exact image-mask-plan mechanism control.
ATConv execution	Checkpoint df18874b ... contains four registered replacements. Forward hooks on a real BUSI test case recorded one call with non-zero output for each projector replacement and zero calls for both neck candidates.	Reported as two forward-active projector operators; no ATConv4 claim.
Completion identity	Run metadata include protocol, manifest, code, cache, recipe, augmentation, epoch, checkpoint, and run-configuration hashes. The completion digest did not include every CLI field, including batch size and worker count; original command logs record them. All 90 checkpoint SHA256 values were independently recomputed and matched.	Hash identity is supplemented by archived commands and disclosed as a limitation.
Source reconstruction	The 90-cell batch records aggregate code digest 2732eeac The recorded Git commit omits four orchestration/evaluation files, and the current, July 15, and July 18 release copies do not reproduce this digest.	The digest identifies a common execution batch, but exact clean-clone source reconstruction is not claimed.
Training-log integrity	Eighty-nine cells contain 100 epoch records with matching declared and selected checkpoint paths. One intermediate BioMedCLIP-ATConv COVID-19 seed-789 log was sparsely overwritten during a duplicate launch; its run metadata and all four controls consistently bind checkpoint 82e42695	The numerical cell remains checkpoint-traceable, while the incomplete training log is not described as pristine provenance.

Exploratory continuous-location diagnostic

This earlier snapshot uses the same transform distribution but a different augmentation seed key from the reference run. It is therefore retained only as a same-distribution diagnostic, not an exact matched mechanism control.

Table S12 Exploratory three-seed continuous-location recomputation diagnostic from an earlier fixed-manifest checkpoint snapshot. MEDEQUISEG + Location Recompute recomputes coarse position phrases from the transformed training mask when a mild affine, translation, or zoom is sampled. Values are mean $\pm$ sample standard deviation. Delta and the reported bootstrap interval are MEDEQUISEG + Location Recompute minus its reference snapshot in percentage points. Both runs use the same transform distribution but different augmentation seed keys, so the realized per-case plans are not exactly matched. These rows are not pooled with the frozen 90-cell ablation. p_W is the exploratory, unadjusted two-sided paired Wilcoxon p -value.

Dataset	Metric	Reference snapshot	Location Recompute	Δ pp [95% CI]	p_W
BUSI	Dice	85.74 $\pm$ 0.80	84.62 $\pm$ 0.20	-1.12 [-2.62, 0.07]	.004
	IoU	79.46 $\pm$ 0.83	78.06 $\pm$ 0.64	-1.40 [-2.70, -0.30]	.002
ClinicDB	Dice	93.18 $\pm$ 0.54	93.06 $\pm$ 0.72	-0.13 [-1.25, 1.27]	.138
	IoU	87.77 $\pm$ 0.82	87.54 $\pm$ 1.07	-0.23 [-1.94, 1.78]	.117

Presence/class-stratified prompt derangement

This post hoc sensitivity analysis reuses the 15 frozen complete-model checkpoints and performs no retraining. For each public dataset, the complete test-prompt multiset is deterministically permuted within target-presence and diagnostic-class strata constructed from released case identifiers and prompt fields. The control builder reads neither image pixels nor masks and validates one-to-one prompt closure, donor identity, stratum membership, and text hashes. It changes 3,413 of 3,426 prompt strings; the 13 unavoidable unchanged cases are BUSI normal studies whose within-stratum strings are identical. Statistics first average the three checkpoint values for each case, then use paired 10,000-repeat bootstrap intervals and Holm-adjusted Wilcoxon tests. The analysis remains a privileged-prompt diagnostic because the strata and prompts encode released target-derived information.

Table S13 Post hoc frozen-checkpoint prompt control that permutes the complete prompt multiset within target-presence and diagnostic-class strata. Changed/N is the number of cases receiving a different text string. Dice values are percentages (mean $\pm$ sample standard deviation across three checkpoints); deltas are case-first true-minus-control effects with 95% bootstrap confidence intervals. Holm p is the adjusted paired Wilcoxon value across the five datasets.

Dataset	Changed/N	True Dice	Stratified	Δ [95% CI]	Holm p
BUSI	65/78	85.78 ± 0.45	79.87 ± 1.05	$+5.91$ [$+2.62, +9.86$]	.038
ClinicDB	61/61	92.07 ± 0.71	88.71 ± 0.37	$+3.37$ [$+0.54, +6.96$]	.142
BUS-BRA	282/282	90.24 ± 0.14	90.16 ± 0.12	$+0.08$ [$-0.10, +0.26$]	.311
BRISC	892/892	87.18 ± 0.14	86.81 ± 0.30	$+0.37$ [$+0.13, +0.65$]	.011
COVID-19	2113/2113	84.08 ± 0.13	65.63 ± 1.10	$+18.45$ [$+17.68, +19.22$]	$< 10^{-6}$
Public-5	3413/3426	87.87 ± 0.17	82.24 ± 0.16	$+5.63$ [$+4.71, +6.67$]	–